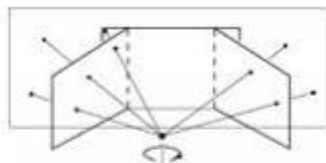
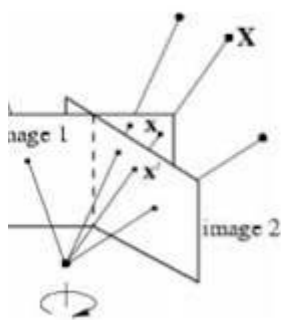




g camera, arbitrary world

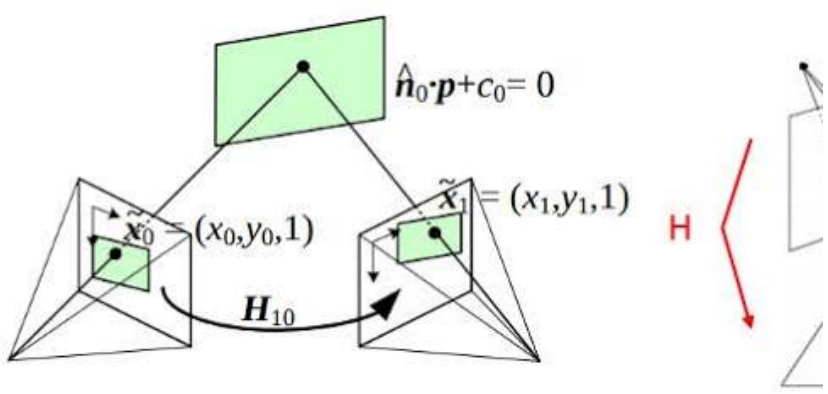
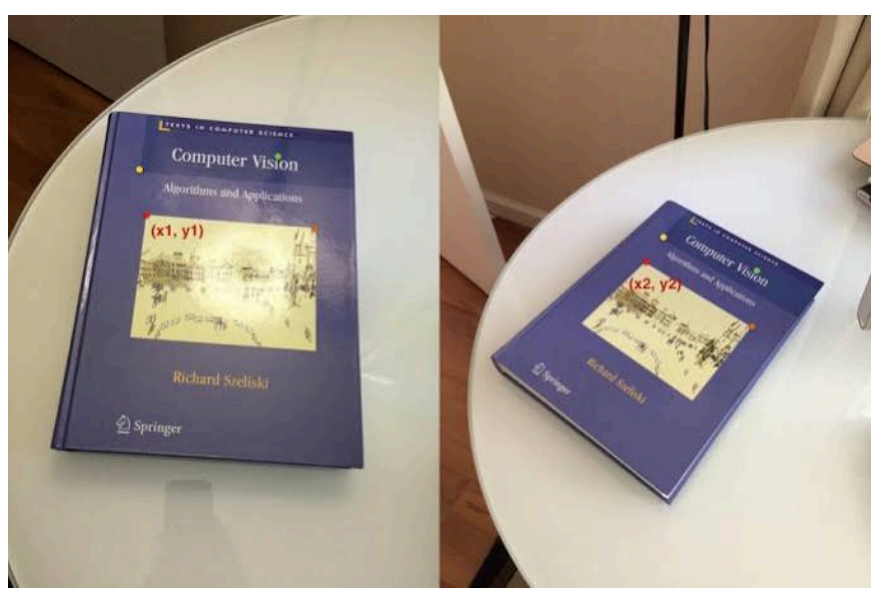
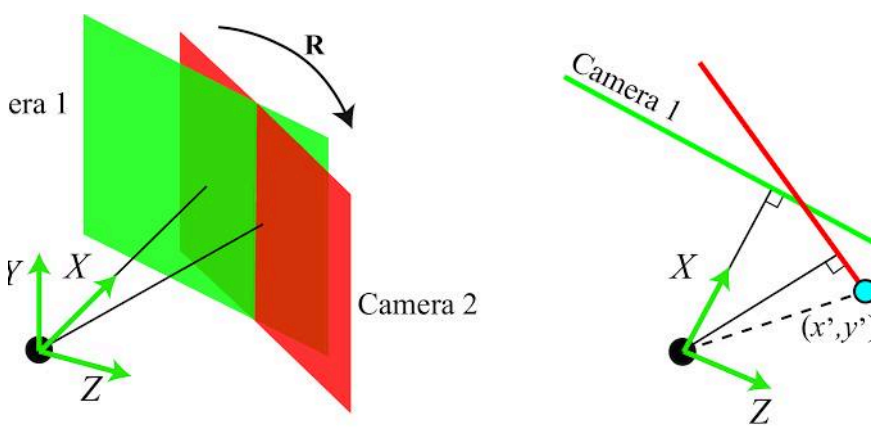
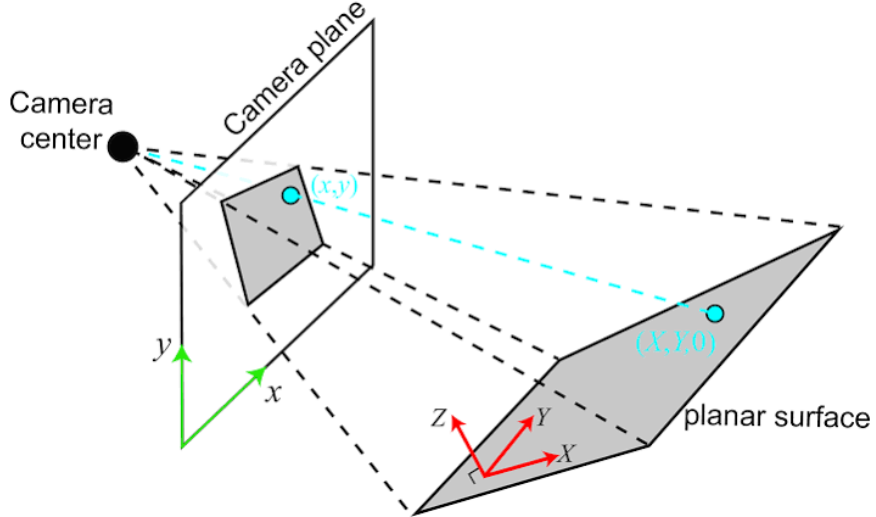


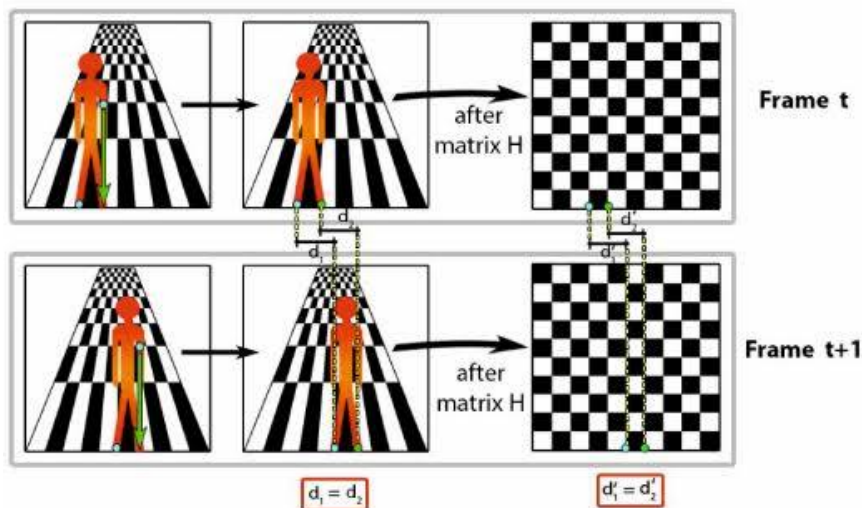
2D transformation hierarchy

A square transforms to:



Projective 8dof	$\begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}$	
Affine 6dof	$\begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$	
Similarity 4dof	$\begin{bmatrix} sr_{11} & sr_{12} & t_x \\ sr_{21} & sr_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$	
Euclidean 3dof	$\begin{bmatrix} r_{11} & r_{12} & t_x \\ r_{21} & r_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$	





In computer vision, a **homography** is a geometric transformation matrix that relates two different planar projections of the exact same 3D surface. Represented as a **3x3 matrix (H) with 8 degrees of freedom**, it maps pixels from a source image to a destination image in homogeneous coordinates. Essentially, if you have two pictures of a flat wall taken from different angles—or two pictures taken from a single camera that only rotated without moving—the pixels between those images are perfectly linked by a homography matrix.

Mathematical Representation

Mathematically, a source point (x_1, y_1) maps to a target point (x_2, y_2) using the transformation matrix:

$$\begin{pmatrix} x_2 \\ y_2 \\ 1 \end{pmatrix} = H \cdot \begin{pmatrix} x_1 \\ y_1 \\ 1 \end{pmatrix} = \begin{pmatrix} h_{00} & h_{01} & h_{02} \\ h_{10} & h_{11} & h_{12} \\ h_{20} & h_{21} & 1 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ y_1 \\ 1 \end{pmatrix}$$

Because the matrix is scale-invariant (defined up to an arbitrary scale factor), the bottom-right value is normalized to 1, leaving **8 independent parameters** to solve.

How Homography is Estimated

To compute a homography matrix, you need a minimum of **4 corresponding point pairs** between the two views:

- **Linear Systems:** Each matching pair provides 2 independent geometric constraints, generating 8 equations to solve the 8 unknown parameters.
- **Algorithms:** It is typically computed using the Direct Linear Transform (DLT) algorithm via Least Squares or Singular Value Decomposition (SVD).
- **Outlier Rejection:** In real-world scenarios, a robust estimator like **RANSAC** (Random Sample Consensus) is applied alongside feature matchers (like SIFT or ORB) to filter out bad pixel matches.
- **Deep Learning Era:** Modern systems also use frameworks like HomographyNet on Emergent Mind to predict displacements directly via supervised or unsupervised neural networks.

Common Applications

- **Image Stitching (Panoramas):** Aligning and warping overlapping photos taken by a rotating camera into a single seamless image.
- **Perspective Correction (Document Scanning):** Straightening an image taken at an angle (like a whiteboard or a receipt) so it looks like it was captured directly from above.
- **Augmented Reality (AR):** Projecting virtual elements (like digital billboards on sports fields or Snapchat filters) onto real-world flat surfaces with perfect perspective.

Are you looking to **implement homography in code** (such as with OpenCV and Python), or do you need help **solving a specific linear algebra equation** for a project?

AI responses may include mistakes. [Learn more about](#)

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The phrase "**The World Eater**" typically points to one of a few iconic pop-culture and fantasy art contexts. Depending on what you are looking for, it most likely refers to one of the following:

1. Francisco Goya-Inspired Warhammer Art

If you are looking for a literal fine-art-style painting, there is a famous and widely shared piece of *Warhammer 40k* fan art by artist Reynier Vorobey. The artwork features a **World Eater Chaos Space Marine** macabrely devouring a victim. The piece is a direct, dark homage to Francisco Goya's legendary black painting, *Saturn Devouring His Son*.

2. Alduin from *The Elder Scrolls V: Skyrim*

In gaming, **Alduin the World-Eater** is the primary dragon antagonist of *Skyrim*. There are several famous official concept paintings and promotional artworks of him—most notably the textured, stone-mural style depictions found on **Alduin's Wall** in-game, as well as digital speedpaintings depicting him unleashing fire over Helgen.

3. Galactus from Marvel Comics

Often titled "The World Eater" or "Devourer of Worlds," Galactus has been the subject of countless iconic comic book cover paintings by master artists like **Alex Ross**, known for his hyper-realistic gouache paintings that bring the cosmic entity to life.

4. Miniature Painting (*Warhammer 40,000*)

If you meant "painting a World Eater," you might be looking for miniature painting guides. The World Eaters are a faction of bloodthirsty space marines whose armor schemes require specific techniques:

- **The Modern Scheme:** A base of deep crimson red (like *Mephiston Red*) contrasted against bright brass or gold trim.
- **The Heresy-Era Scheme:** The classic pre-treason look, which uses a clean off-white base armor with stark blue shoulder pads and accents.

Which of these "World Eaters" were you hoping to find? If it's a specific piece of art, sharing a few details about **what the image looks like** can help pinpoint the exact artist.

AI responses may include mistakes. [Learn more about](#)

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